

A Leakage-Aware, ERA5-Augmented Ensemble for Multi-Horizon Cyclone Track Prediction over the Bay of Bengal

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Abstract—Tropical cyclone track guidance over the Bay of Bengal underpins coastal early warning where dense population and limited supercomputing capacity motivate lightweight predictors fused with Earth-observation-based atmospheric state. We present SECE, a multi-horizon (3–48 h) ensemble that couples IBTrACS kinematics with collocated ERA5 point reanalysis through a dedicated “environ” expert and validation-only fusion routing. Eight tree-based systems are compared under storm-wise splits and a two-tier seed protocol that corrects test-set leakage from iterative architecture tuning. On nine held-out seeds (442 unique test storm IDs pooled), SECE achieves pooled medians of 4.90 km (3 h), 35.54 km (12 h), 101.19 km (24 h), and 246.33 km (48 h). Paired Wilcoxon tests with Bonferroni correction ($\alpha/7$) show SECE significantly outperforms all seven baselines at 3 and 12 h, matches leading stacks at 24–48 h, and never ranks significantly below the field. Controlled ablations attribute incremental skill to point ERA5 features and the environ expert; spatial patch statistics were tested and not adopted. A Cyclone Laila (2010) case study illustrates horizon-dependent ERA5 benefit consistent with aggregate statistics.

Index Terms—Tropical cyclone, track forecast, ERA5, ensemble learning, Bay of Bengal, IBTrACS, disaster early warning

I. INTRODUCTION

The Bay of Bengal and its densely populated rim experience recurrent landfalling tropical cyclones, with track uncertainty driving evacuation lead time and surge exposure [1], [2]. Operational guidance relies on dynamical models and consensus aids [3], [4], yet reanalysis and best-track archives now enable learned predictors that ingest satellite-era environmental fields without running a full NWP cycle [5]–[7].

Machine-learning track models span statistical analogs, gradient-boosted trees, and deep sequence or convolutional encoders over gridded fields [8]–[12]. Recent evidence suggests strong tree ensembles remain competitive on tabular storm–environment tables relative to deep networks [13], while reanalysis winds, pressure, sea-surface temperature (SST), steering flow, and shear remain standard predictors linking EO products to motion [14]–[16]. Basin-specific benchmarks are still sparse for the North Indian Ocean compared with Atlantic training corpora [17], [18].

We address three needs: (i) multi-horizon evaluation at 3, 12, 24, and 48 h on a modern genesis window (1940–2024); (ii) transparent fusion of ERA5 through an interpretable expert channel rather than opaque end-to-end grids alone; and

(iii) rigorous validation that separates architecture development from confirmatory testing—a distinction often blurred when researchers tune ensembles using test-set rank tables [19], [20].

Contributions. (1) An eight-system benchmark with pooled bootstrap confidence intervals and paired significance tests. (2) Subset-Expert Context-Aware Ensemble (SECE v2 Phase 3) (Phase 3), locking 63 inputs (49 track/physics + 14 ERA5 point features). (3) Documentation and correction of a leakage mode from our own development workflow. (4) Negative results on spatial ERA5 patches ($5^\circ/3^\circ$) for reproducibility.

II. DATA AND FEATURE DESIGN

A. Track archive and sampling

We use IBTrACS v4 North Indian Ocean best tracks [6], [18]. Genesis-year filtering 1940–2024 yields 852 storms and 27,728 quality-controlled 3-hourly origins; 583 storms supply 15,605 multi-horizon labels with sufficient forward track length. Storm-wise 70:15:15 train/validation/test partitions are redrawn per random seed ($\approx 408/87/88$ storms per split). Targets are great-circle displacements to the verified position at each lead time.

B. ERA5 collocation

Atmospheric state is taken from ERA5 [5] over $30\text{--}5^\circ\text{N}$, $70\text{--}102^\circ\text{E}$ at each track time. Fourteen point features include 850/500/200 hPa wind components, mean sea level pressure, SST (with a coastal land-mask imputation flag), deep-layer steering (u, v), and vertical wind shear (u, v , magnitude)—quantities routinely derived from satellite-assimilated reanalysis and relevant to Bay of Bengal monsoon steering [15]–[17].

C. Track–physics block

Forty-nine kinematic and derived predictors capture position lags, translation speed and heading, seasonality, recurvature proxies, and land-interaction indicators following persistence and statistical-dynamical baselines [3], [8], [9], [30].

D. Date-window control

We compared genesis windows 1940–2024, 1979–2024, and 1990–2024 under a fixed test draw; 1940–2024 retained long-lead training diversity while acknowledging pre-1979 wind sparsity in IBTrACS [7], [21].

III. VALIDATION PROTOCOL AND LEAKAGE CONTROL

Initial SECE phases were selected while inspecting test-set error ranks—a subtle leakage pathway because base learners never saw test labels but the *architecture* did [?], [20]. We froze Phase 3 after nine *development* seeds used only for fusion-path and subset decisions, then evaluated nine independent *held-out* seeds once without further changes [20]. Development and held-out tier means align closely (e.g., 48 h SECE ≈ 247 km in both), supporting stability rather than split-specific overfitting [19], [22].

Significance analysis pools test predictions across held-out seeds; 442 denotes distinct storm IDs in that pooled set. Primary metric: sample median Haversine error (km). We report 10,000 storm-bootstrap 95% confidence intervals (CI) and paired Wilcoxon signed-rank tests on per-storm medians versus SECE, with Bonferroni threshold $\alpha/7 \approx 0.0071$ across seven baseline comparisons.

IV. METHODS

A. Baseline systems

Seven comparators mirror common operational-statistical and ML practice: Stacking Ensemble (RF+XGB+LightGBM with Ridge meta-learner), Persistence Residual Cascade (PRC) [8], Random Forest, LightGBM, XGBoost, CatBoost, and CatBoost plus a motion neural-network residual (CB+MotionNN). Tree hyperparameters are uniform (300 estimators, max depth 6, learning rate 0.01) to isolate architecture [13], [23]. Deep sequence models were explored in development but omitted from the locked held-out confirm to bound compute and maintain a purely tree-based comparison table.

B. SECE Phase 3

Fig. 1 summarizes the locked pipeline. For each horizon, four feature subsets are trained: at 3 h, position, motion, environ, and full; at 12–48 h, motion, physics, environ, and full. Each subset trains CatBoost, XGBoost, LightGBM, and Random Forest; predictions fuse via non-negative least squares (NNLS) per subset. Subset-NNLS signals join the seven champion baselines in horizon-specific candidate pools (degree-NNLS, km-NNLS, stacking-residual at 3 h). A validation-only router picks the candidate with lowest validation median error per seed; test predictions use that frozen choice.

C. Ablations (development tier)

Track-only SECE, point ERA5 augmentation, dedicated environ expert, and $5^\circ/3^\circ$ spatial patch statistics were compared on development seeds only. Spatial patches improved short-range rank frequency but not 24–48 h means and were rejected for lock—reported as a negative EO-context result relevant to neighborhood versus point fusion studies [10], [24].

V. RESULTS

A. Held-out confirm

Table I lists pooled medians and CIs for all eight evaluated systems. At 3 h, SECE (4.895 km) edges Random Forest (4.927 km) and Stacking (4.945 km); Wilcoxon tests

reject equality versus all seven baselines after Bonferroni correction. At 12 h, SECE (35.538 km) leads Stacking (36.146 km) with the same significance pattern. At 24 h, Stacking (100.771 km) and XGBoost (101.029 km) slightly lead SECE (101.186 km) within overlapping CIs; only CatBoost and CB+MotionNN differ significantly from SECE. At 48 h, XGBoost (242.702 km) and PRC (243.229 km) sit at the top of the ensemble pack; SECE (246.331 km) is statistically tied with Stacking/LightGBM and significantly better than CatBoost only. SECE is never significantly *worse* than any baseline at any horizon among evaluated systems.

Fig. 2 shows the 3 h error–accuracy landscape for all eight systems. Development ablations: track-only SECE averaged 250.3 km at 48 h (mean over dev seeds); adding point ERA5 reduced this to 246.8 km (8/9 seeds improved); the environ expert improved 24 h rank stability versus distributing ERA5 across legacy groups.

B. Diagnostic and case study

A track-only versus ERA5 LightGBM diagnostic on development seed 0 ([13] style tabular comparison) shows ERA5 benefit growing with lead: near parity at 24 h ($\approx 49\%$ of test storms improved), majority benefit at 48 h ($\approx 61\%$, median ≈ 11 km). Cyclone Laila (2010) in Fig. 3 illustrates large single-storm ERA5 gain in that diagnostic while aggregate 48 h rankings remain compressed—consistent with ensemble parity at long range [12], [22]. On held-out seed 0, SECE’s Laila 24 h per-storm median is ≈ 111 km. A second held-out storm (1996, central-basin genesis) shows tighter 24 h contest between SECE and Stacking (≈ 107 vs. 111 km) and wider 48 h separation favoring XGBoost—mirroring Table I rather than case-specific ranking.

Table II summarizes Bonferroni-corrected Wilcoxon outcomes versus SECE (seven baselines $\times$ four horizons). No competitor is significantly *better* than SECE at any horizon; SECE is significantly *better* than all seven at 3–12 h.

C. Reproducibility

All locked held-out prediction CSVs, bootstrap CI tables, and case-study trajectory exports are versioned with the “NEW WAY” pipeline; figures regenerate from a single script using Cartopy basemaps. Held-out evaluation required ≈ 9.4 h on a workstation (Intel i7-13700HX, 16 GB RAM); development ablations ranged 4.9–10.9 h per configuration. Deep learning baselines (CNN-GRU, BLSTM) were trained in exploratory phases but excluded from the confirm table to keep the eight-system tree benchmark closed under fixed compute [10], [11], [29].

VI. DISCUSSION AND CONCLUSION

Point ERA5 features provide measurable incremental skill without gridded encoders in the locked design, aligning with EO–ML workflows that colocate reanalysis or satellite winds with storm centers [14], [16], [25]. SECE’s routed subset experts help at 3–12 h where local steering and land cues matter; at 24–48 h, simple stacks remain strong, echoing

TABLE I
HELD-OUT PERFORMANCE: MEDIAN GREAT-CIRCLE ERROR (KM) WITH 95% BOOTSTRAP CI (NINE SEEDS POOLED; 442 STORM IDS). CNN/BLSTM NOT IN LOCKED CONFIRM.

Horizon	System	Median	CI low	CI high
3 h	SECE	4.895	4.714	5.079
	Random Forest	4.927	4.734	5.125
	Stacking	4.945	4.763	5.123
	XGBoost	6.411	6.243	6.580
12 h	SECE	35.538	34.208	36.925
	Stacking	36.146	34.744	37.701
	LightGBM	36.928	35.498	38.338
	CatBoost	42.028	40.381	43.656
24 h	Stacking	100.771	96.887	104.703
	SECE	101.186	97.301	104.808
	PRC	101.941	98.131	105.824
	CatBoost	107.191	102.555	111.839
48 h	XGBoost	242.702	232.596	255.403
	Stacking	245.977	235.938	257.183
	SECE	246.331	235.541	257.614
	CatBoost	254.862	242.991	267.797

TABLE II
WILCOXON VS. SECE AFTER BONFERRONI ($p < 0.0071$): + = SECE BETTER; 0 = NOT SIGNIFICANT; - = BASELINE BETTER (NONE OBSERVED).

Baseline	3 h	12 h	24 h	48 h
Stacking	+	+	0	0
Random Forest	+	+	0	0
LightGBM	+	+	0	0
XGBoost	+	+	0	0
PRC	+	+	0	0
CatBoost	+	+	+	+
CB+MotionNN	+	+	+	0

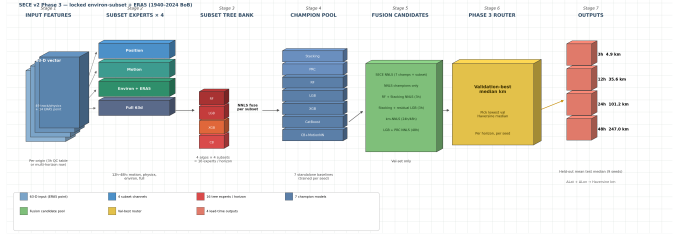


Fig. 1. Locked SECE Phase 3: subset tree banks, environ channel (14 ERA5 features), validation router.

broader predictability limits for synoptic-scale track error [3], [22]. Honest reporting of non-significant long-range separations and rejected spatial patches is intended to anchor future IGARSS-relevant work on satellite wind assimilation, probabilistic cones, and physics-informed neural residuals [11], [19], [26], [27]. From a disaster-management perspective, the 3–12 h gains are most actionable for port and shelter decisions, whereas 24–48 h parity implies that consensus-style stacks remain difficult to beat without new spatial EO encoders or assimilation-aware features [1], [3], [25]. Statistical-dynamical hybrids such as SHIPS-class environmental predictors motivate our point-feature design even though SECE targets track rather than intensity [3], [9].

We release datasets, held-out prediction tables, and figure scripts as a reproducible Bay of Bengal benchmark. Future work includes gridded CNN/Transformer encoders, coupling with scatterometer winds, and extension to intensity and rainfall for integrated early warning [1], [28], [29].

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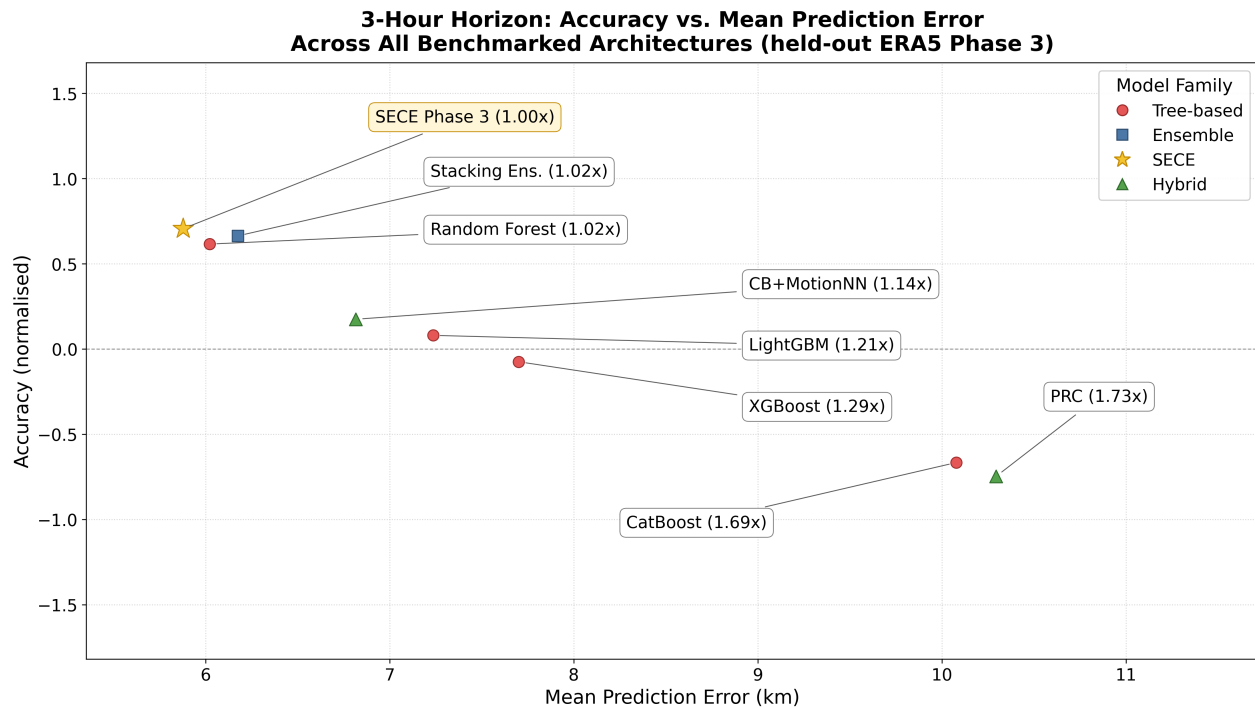


Fig. 2. Held-out 3 h benchmark: mean error versus normalized accuracy for eight systems (locked ERA5 environ-subset Phase 3). Leader-line labels match the legacy visualization used in our journal draft.

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Trajectory Validation — LAILA (2010): Held-out seed 0 vs IBTrACS

